Model Paper - Database Management Systems

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Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)

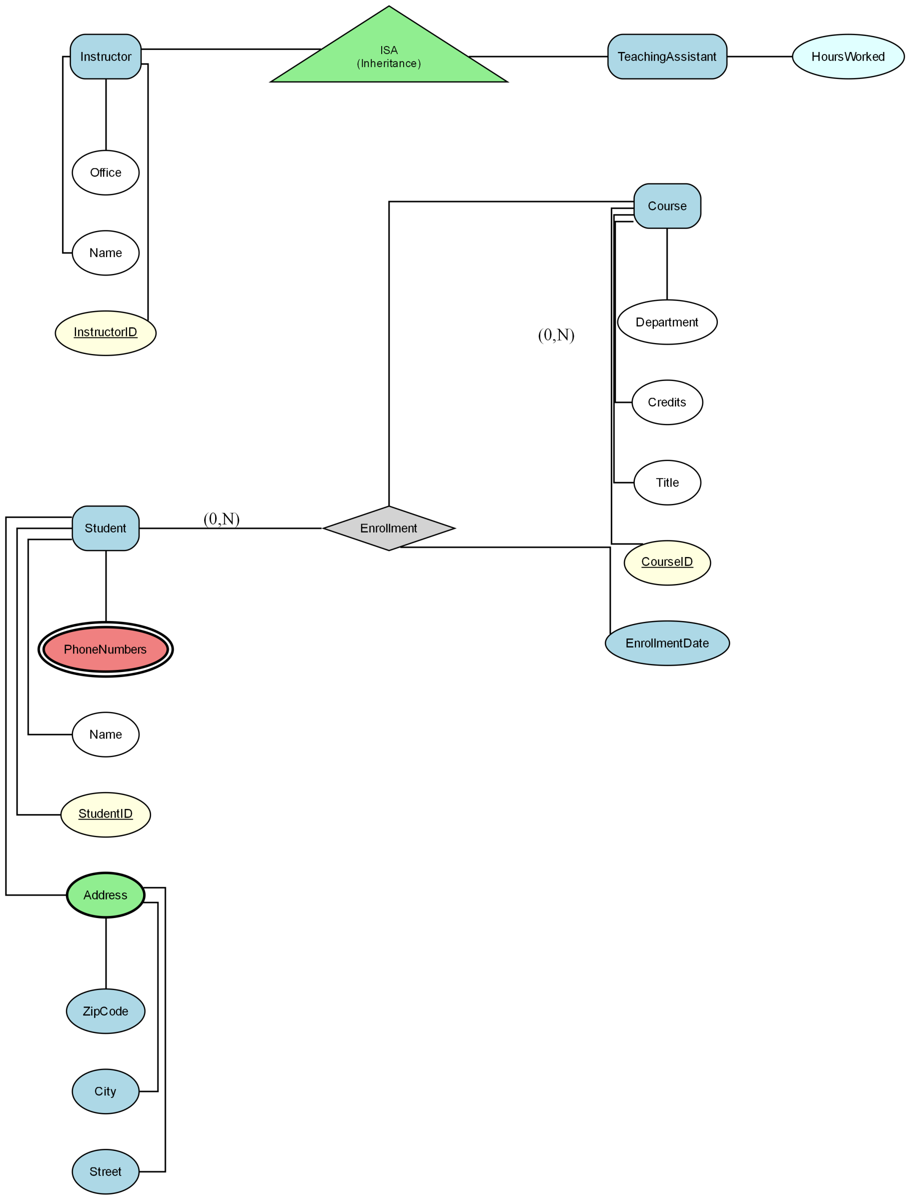
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)

- (0,1) indicates optional participation (zero or one)

- (0,N) or (0,*) indicates optional many participation (zero or many)

A university manages its students and courses through an Entity-Relationship model. The 'Student' entity has attributes: StudentID, Name, Address, and PhoneNumbers. The 'Course' entity includes CourseID, Title, Credits, and Department. There is also an 'Enrollment' relationship between 'Student' and 'Course', which has a descriptive attribute EnrollmentDate. In addition, there is an 'Instructor' entity with attributes: InstructorID, Name, and Office. An 'ISA' relationship defines that 'TeachingAssistant' is a subtype of 'Instructor', and it has a specific attribute: HoursWorked. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrollment' relationship (Student participation is optional, Course participation is optional)

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation R(A, B, C, D, E) with the following set of functional dependencies F over R: F={{AB->C, A->D, C->E, B->CD}}.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A healthcare organization is developing a database system to manage its patients' medical records. Different roles are assigned to various staff members for database administration and management. Emily is the senior database administrator responsible for overseeing the database system's creation and maintenance. Alex and Jordan are junior database administrators working under Emily's supervision. Alex manages the patient records database, while Jordan handles the appointments database. Sam is a database developer reporting to Alex and Jordan. His responsibilities include designing and implementing database schemas for patient records and appointments. Lisa and Mark are data entry operators responsible for inputting patient information and appointment details into the system.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: ```java String sql = "SELECT * FROM employees WHERE department = ?"; PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs = pstmt.executeQuery(); ``` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2 marks)

d) Briefly explain how authentication and authorization are achieved in SQL Server. (3 marks)

e) A hospital is establishing role-based access control for its database system that manages Patients, Doctors, and Appointments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Appointments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating patient appointment data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Loans database using a user-defined server role. (5 marks)
iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Members and Loans databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a university: Student (studentId: int, name: varchar(100), email: varchar(50), phone: varchar(15)) Course (courseId: int, title: varchar(100), description: varchar(250), credits: int) Enrollment (enrollmentId: int, studentId: int, courseId: int, enrollmentDate: date, grade: varchar(2)) The 'Student' table stores information about students enrolled in the university, including unique student IDs, names, email addresses, and phone numbers. The 'Course' table contains details about courses offered, including unique course IDs, titles, descriptions, and credit hours. The 'Enrollment' table records the relationship between students and courses, detailing which students are enrolled in which courses, along with their enrollment dates and grades. The 'Enrollment' table manages student course enrollments, recording which students are enrolled in which courses and when. Fee (feeId: int, studentId: int, amount: real, paymentStatus: varchar(20)) The 'Fee' table stores fee/payment details including payable amount and payment status.

a) Write SQL Queries to perform the following: i. Find the names and emails of students enrolled in the course titled 'Database Systems'.

i. Find the names and emails of students enrolled in the course titled 'Database Systems'. (4 marks)
ii. Find the student with the highest aggregate value using GROUP BY on related enrollment records. (6 marks)
iii. Find the title and name from Course, Student, and Enrollment where Enrollment.enrollmentId is not null. (7 marks)

b) b) Create a function to calculate the total fee amount for a given student. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each student. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by student. The student pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)